Chapter-4

# *Rotational Kinematics and Conservation of Angular momentum*

***Textbook: Fundamentals of Physics by Halliday, Resnick & Walker [extended 10th edition]***

## *Topics to be Covered:*

10-2: Rotation with Constant Angular Acceleration. 10-3: Relating the Linear and Angular Variables. 10-4: Kinetic Energy of Rotation. 10-5: Calculating the Rotational Inertia, 10-6: Torque. 11-5: Angular Momentum. 11-8: Conservation of Angular Momentum.

## *Theory*

1. Explain the terms (i) moment of inertia (ii) radius of gyration (iii) parallel axes theorem (iv) Torque.
2. Derive the expression for kinetic energy of rotation.
3. Calculate the rotational inertia of a thin rod about the central axis of rotation perpendicular to length.
4. Show that $\tau = I\alpha$.
5. What is angular momentum? Explain the law of conservation of angular momentum.

## *Rotational Kinematics (Chapter-10)*

1. [5] A diver makes 2.5 revolutions on the way from a 10-m high platform to the water. Assuming zero initial vertical velocity, find the average angular velocity during the dive.

2. [9] A drum rotates around its central axis at an angular velocity of 12.60 rad/s. If the drum then slows at a constant rate of 4.20 rad/s$^2$, through what angle does it rotate in coming to rest?

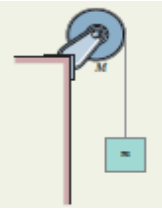
3. [16] A merry-go-round rotates from rest with an angular acceleration of 1.50 rad/s$^2$. How long does it take to rotate through the first 2.00 rev?

4. [19] What are the magnitudes of (a) the angular velocity, (b) the radial acceleration, and (c) the tangential acceleration of a spaceship taking a circular turn of radius 3220 km at a speed of 29000 km/h?

5. [23]A flywheel with a diameter of 1.20 m is rotating at an angular speed of 200 rev/min. (a) What is the angular speed of the flywheel in radians per second? (b) What is the linear speed of a point on the rim of the flywheel? (c) What constant angular acceleration (in revolutions per minute-squared) will increase the wheel's angular speed to 1000 rev/min in 60.0 s? (d) How many revolutions does the wheel make during that 60.0 s?

6. [22] An astronaut is being tested in a centrifuge. The centrifuge has a radius of 10 m and, in starting, rotates according to $\theta = 0.30t^2$, where t is in seconds and u is in radians. When t = 5.0 s, what are the magnitudes of the astronaut's (a) angular velocity, (b) linear velocity, and (c) tangential acceleration?

7. [91] In Figure, a wheel of radius 0.20 m is mounted on a frictionless horizontal axis. The rotational inertia of the wheel about the axis is 0.40 kg $m^2$. A massless cord wrapped around the wheel's circumference is attached to a 6.0 kg box. The system is released from rest. When the box has a kinetic energy of 6.0 J, what is the wheel's rotational kinetic energy?



## *Conservation of Angular Momentum (Chapter-11)*

8. [46] The rotational inertia of a collapsing spinning star drops to $\frac{1}{3}$ its initial value. What is the ratio of the new rotational kinetic energy to the initial rotational kinetic energy?

9. [27] At one instant, force $\vec{F} = 4.0\widehat{jN}$ acts on a 0.25 kg object that has position vector $\vec{r} = (2.0\hat{\imath} - 2.0\hat{k})m$ and velocity vector. $\vec{v} = (-5.0\hat{\imath} + 5.0\hat{k})m/s$. About the origin and in unit vector notation, what are (a) the object's angular momentum and (b) the torque acting on the object?

10. [74] A wheel rotates clockwise about its central axis with an angular momentum of 600 kg .$m^2$/s. At time t = 0, a torque of magnitude 50 N.m is applied to the wheel to reverse the rotation. At what time t is the angular speed zero?

11. [53] In the figure, a uniform thin rod of length 0.500 m and mass 4.00 kg can rotate in a horizontal plane about a vertical axis through its center. The rod is at rest when a 3.00 g bullet traveling in the rotation plane is fired into one end of the rod. In the view from above, the bullet's path makes angle $\theta = 60^0$ with the rod. If the bullet lodges in the rod and the angular velocity of the rod is 10 rad/s immediately after the collision, what is the bullet's speed just before impact?

